

Discussion Rouleaux Mercredi 22/04/2026

Laboratoire d'Aérodynamique (LAERO) · Université de Toulouse

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But de la réu :

- Présenter ma méthode et questionnements
- Échange d'idées ? venant de vos expériences pour débloquer et/ou détecter fausses routes

Quels processus physiques derrières les rouleaux?

Différentes instabilités (Etling and Brown 1993):

- Instabilité point d'inflexion (theorème de Rayleigh, Brown 1980)
- Instabilité parallèles, pour un fluide en rotation
- **Instabilité thermique**

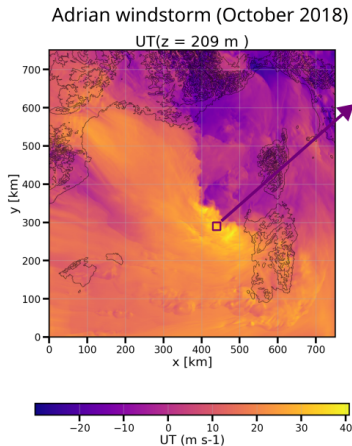
Des processus ont surement une influence:

evaporation et sublimation (Rivière, Ricard, and Arbogast 2020) ; hétérogénéités de surface (M. Gryschka et al. 2008)

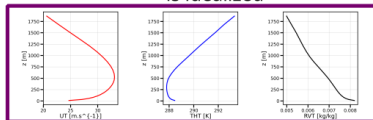
Quelle question scientifique ?

- Comprendre les processus qui mène à la formation de rouleaux, quelles instabilité ?
- Rôle dans le transport de quantité de mouvement, humidité et chaleur ?
Dans les CAO rôle possiblement négligeable (Micha Gryschka, Fricke, and Raasch 2014)
- Vents forts : problématique des flux de surfaces, possibilité de liens entre les flux de surface et la géométrie des rouleaux ? Lfarh et al. 2024

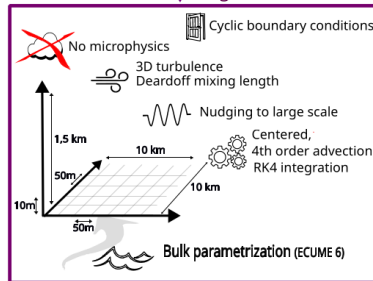
Quelle méthode pour étudier les rouleaux?



The atmospheric profile in the boundary layer is idealized



used as initial and large scale condition



Nudging:

Addition of a force in order to account for the large-scale forcings acting on the variables within the simulation domain. (tau = 200s)

$$\frac{\partial \phi}{\partial t}(z) = \frac{1}{\tau} (\langle \phi_{LS}(z) \rangle - \langle \phi(z) \rangle)$$

Objectives:

To reproduce the rolls obtained in large-eddy simulations (Lfhar et al. 2024)

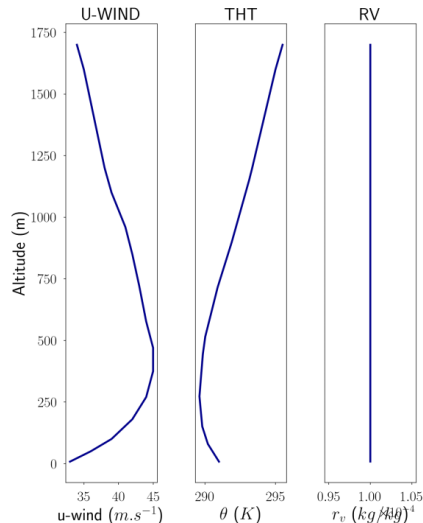
To study the properties of rollers under modified large-scale conditions

Quels diagnostics sont faits ?

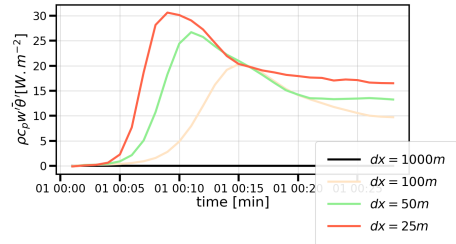
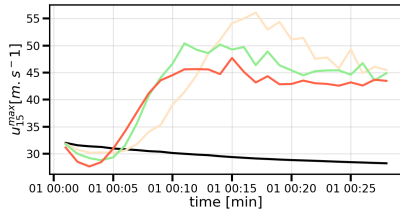
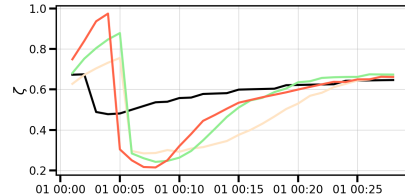
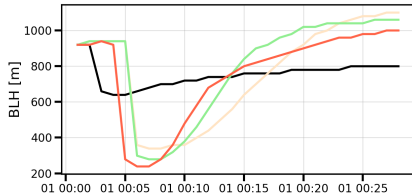
- **Hauteur de couche limite** diagnostique online
Choix sur la KE, si $< 5\%$ de la KE au dessous ?
Alternatives sont : sur le gradient de température / sur évolution TKE / flux wthv
- **Longueur de Moonin Obhukov** online:
- Paramètre de stabilité $\zeta = z_i/L$
 $6 < \zeta < 15$ (Etling and Brown 1993) dans le cas d'instabilité thermique

Simulation ADRIAN_3D - large domain

Prep ideal	Domain size discretization	25 x 25 x 1.6 km $dx = dy = 1000 - 100 - 50 - 25m$ $10m \leq dz \leq 20m$
	Boundaries	2* cyclic, no slip
mesonh	Schemes Mixing length Integrat° step Integrat° length Advection Temporal Relaxation τ^{relax}	TKEL, REVE, PPM DEARDOFF varie avec dx s 1700 s CEN4TH RK4 $\bar{U}$, $\bar{\Theta}$ 2000 s
profiles	θ_g sst	291K 295K

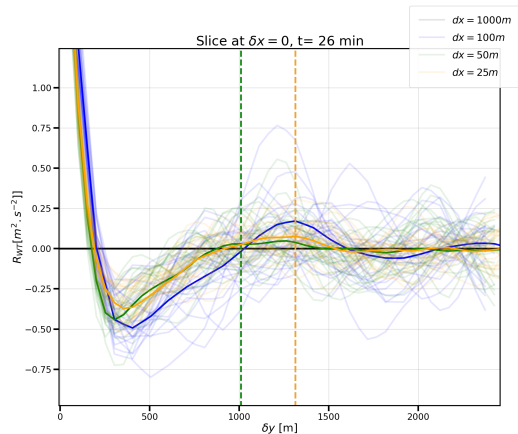
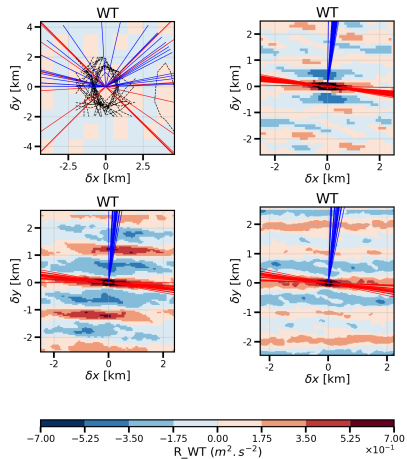


Evolution



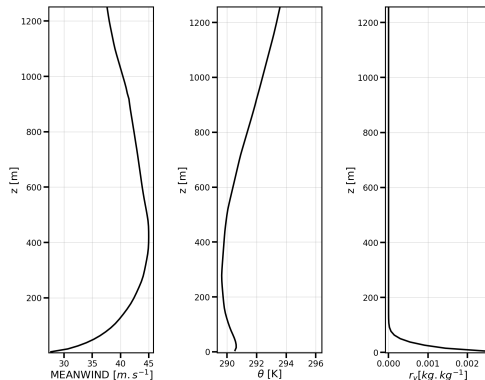
$$\text{ACF } R(\delta x) = \langle W(x + \delta x) \times W(x) \rangle$$

WT at $t = 26$ min; $z = 259$ m

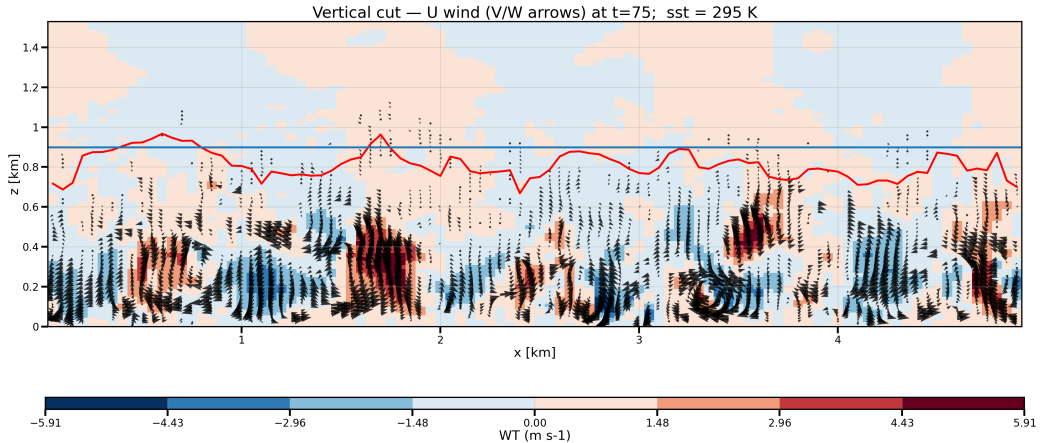


Simulation ADRIAN_3D

Prep ideal	Domain size discretization	$5 \times 5 \times 1.6 \text{ km}$ $dx = dy = 50m$ $10m \leq dz \leq 20m$
	Boundaries	2* cyclic, no slip
mesonh	Schemes Mixing length Integrat° step Integrat° length Advection Temporal Relaxation τ^{relax}	TKEL, PPM DEARDOFF 0.5 s 5000 s CEN4TH RK4 $\bar{U}, \bar{V}, \bar{\Theta}$ 60 s
profiles	θ_g sst	291K 295 K

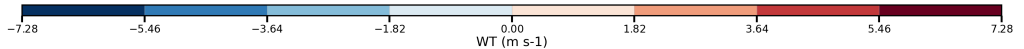
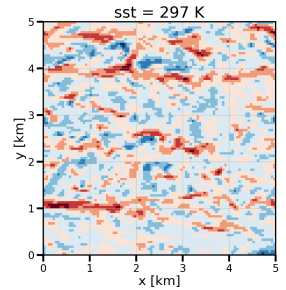
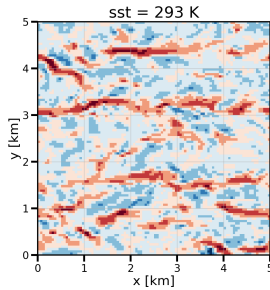
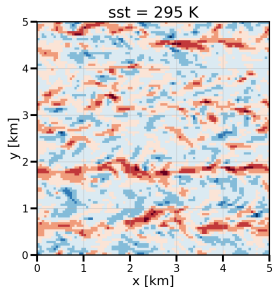


Coupe verticale

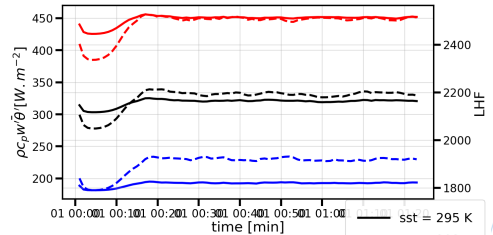
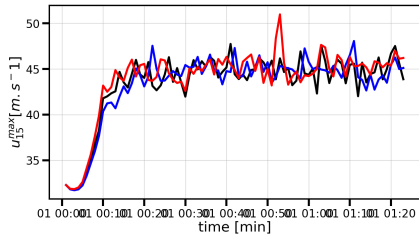
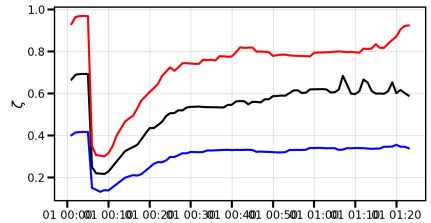
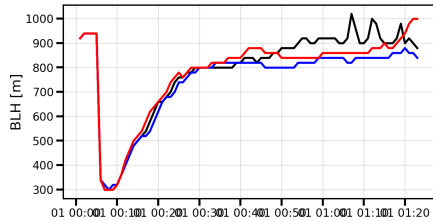


Comparaison plusieurs sst

WT; level=259 m; t = 75 min

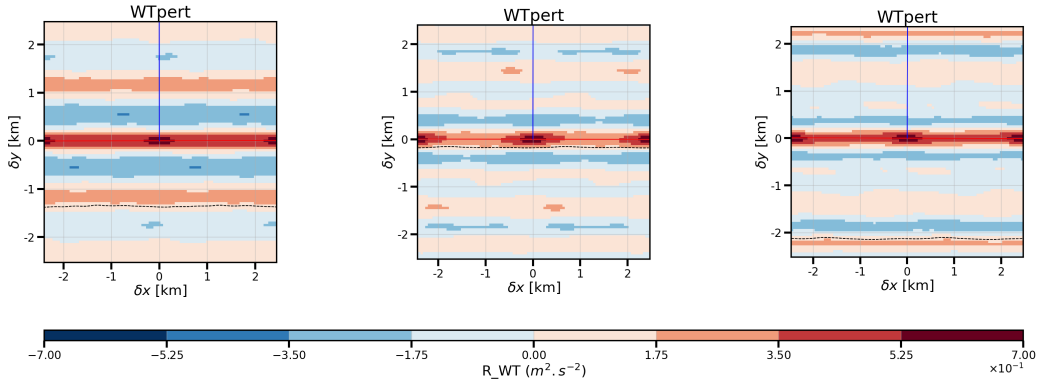


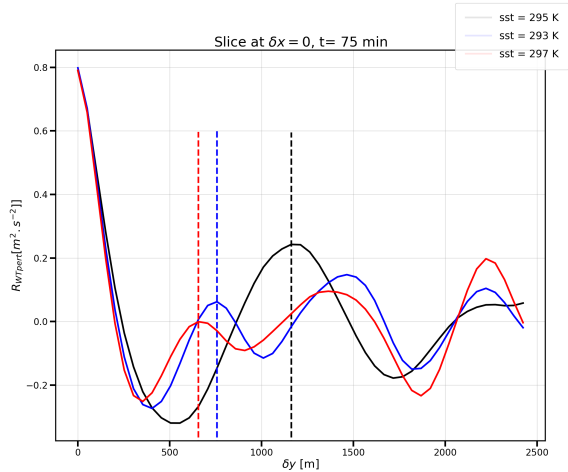
Evolution



Evolution

$\bar{W}_t - \langle \bar{W}_t \rangle$ on [70, 80] min; $z = 259$ m





Quelles limites et pb ?



Etling, D. and R. A. Brown (Aug. 1993). “Roll Vortices in the Planetary Boundary Layer: A Review”. In: *Boundary-Layer Meteorology* 65.3, pp. 215–248. ISSN: 0006-8314, 1573-1472. DOI: 10.1007/BF00705527. URL: <http://link.springer.com/10.1007/BF00705527> (visited on 09/23/2025).



Gryschka, M. et al. (Dec. 2008). “On the Influence of Sea-ice Inhomogeneities onto Roll Convection in Cold-air Outbreaks”. In: *Geophysical Research Letters* 35.23, 2008GL035845. ISSN: 0094-8276, 1944-8007. DOI: 10.1029/2008GL035845. URL: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2008GL035845> (visited on 11/19/2025).



Gryschka, Micha, Jens Fricke, and Siegfried Raasch (Nov. 27, 2014). “On the Impact of Forced Roll Convection on Vertical Turbulent Transport in Cold Air Outbreaks”. In: *Journal of Geophysical Research: Atmospheres* 119.22. ISSN: 2169-897X, 2169-8996. DOI: 10.1002/2014JD022160. URL: <https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2014JD022160> (visited on 11/18/2025).